Integrating Recurrent-KAN With SAM Adapter for Blind Hyperspectral Unmixing

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***Abstract*—Due to the limitation of sensors, hyperspectral images (HSIs) contain a large number of mixed pixels. Hyperspectral unmixing (HU) techniques decompose these mixed pixels into distinct end-members and their corresponding abundance values. Traditional methods initialize weights in the decoder and utilize outputs/weights as abundance maps and end-members—an approach heavily dependent on initial weights that significantly limits performance. This article proposes a blind HU method integrating recurrent Kolmogorov–Arnold networks (KANs) with segment anything model (SAM) adapter. The method operates through three sequential stages: feature encoding, end-member extraction, and abundance estimation. Specifically for feature encoding, an HU-SAM adapter is proposed to capture global–local spatial features. For end-member extraction, an iteratively learned Recurrent-KAN module reconstructs end-members while stabilizing model learning. For abundance estimation, an updated Swin Transformer module is utilized to maintain a lower parameter count. Extensive experiments on real and synthetic datasets demonstrate superior effectiveness of the proposed method over the eight state-of-the-art methods.**

***Index Terms*—Blind hyperspectral unmixing (HU), Kolmogorov–Arnold networks (KANs), linear spectral mixture model (LSMM), segment anything model (SAM).**

I. Introduction

HYPERSPECTRAL images (HSIs) capture spectral information across hundreds of contiguous wavelengths, enabling precise material identification and discrimination for diverse applications [1], [2], [3]. However, constrained by sensor limitations and complex surface distributions, HSIs frequently contain abundant mixed pixels that hinder distinct material separation. In hyperspectral unmixing (HU), end-members within a pixel are typically assumed to be linearly or nonlinearly mixed [4], [5], [6]. HU addresses this by decomposing mixed pixels into end-members and their corresponding abundance values at each pixel, thereby providing spectral signatures and spatial distribution of constituent materials.

For the HU task, the methods can be roughly divided into two categories: nonlinear spectral mixture model (NSMM) and linear spectral mixture model (LSMM) [7], [8], [9]. NSMMs account for multiple scattering effects among materials, thereby providing a more realistic characterization of the imaging mechanism [10], [11], [12]. Typically, NSMMs can be unified under two paradigms—additive nonlinear and postnonlinear—and further organized into a hierarchical family of models [13]. Bilinear [14] and generalized bilinear models [15] characterize secondary scattering; postnonlinear and additive postnonlinear models extend the nonlinear mapping capacity [16]; multilinear (MLM) [17] and p-linear models [18], [19], [20] are able to describe higher order interactions; kernel-based approaches [21], [22] leverage reproducing kernel Hilbert space mappings to linearize nonlinear mixtures; and radiative transfer models [23], [24] approximate intimate mixtures by linearization in the single-scattering albedo domain. Despite their physical interpretability, NSMMs suffer from several limitations. Bilinear-type models can only capture restricted forms of nonlinearity and often underperform in complex real-world scenarios [25]. MLM and p-linear models rapidly expand the parameter space as the order p increases, leading to overfitting risks. Radiative transfer models require scene-specific parameters that are difficult to estimate, limiting generalization across different datasets. Kernel methods depend heavily on kernel function selection, parameter tuning, and often require prior scene knowledge [25]. Furthermore, NSMMs generally incur high computational costs and exhibit strong sensitivity to parameter initialization, making them far less efficient than LSMMs [26].

In contrast, LSMMs assume that each mixed pixel can be expressed as a linear combination of end-members [13], [27], subject to the abundance non-negativity and sum-to-one constraints. Prior-based methods [28], [29] estimate abundances given known pure end-members or spectral libraries, but accurate end-member information is often unavailable in

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practice. Classical matrix factorization and geometric-based methods [30], [31], [32], [33] relax this requirement by jointly estimating end-members and abundances without prior knowledge; however, they are highly sensitive to initialization and typically involve nonconvex optimization problems. Nevertheless, compared with NSMMs, LSMMs remain central in HU research and applications owing to their simplicity, flexibility, and computational efficiency.

Recent years have witnessed significant advances in deep learning (DL)-based HU methods [25], [34], [35], [36], leveraging the powerful learning and data-fitting capabilities of deep neural networks. Zhao et al. [37] employed stacked 3-D convolutional layers and structured sparse regularization to enforce sparsity in abundance estimation. Wang et al. [38] utilized Swin Transformer modules to enhance both local and nonlocal spatial information. Yang et al. [39] adopted a dual-branch architecture to separately capture spectral and spatial features of HSIs. Dong et al. [40] built a superpixel-based graph network that integrates autoencoders for local feature extraction and introduces regularization to enforce sparse and low-rank abundance estimation. Qi et al. [41] proposed a spatial attention-weighted unmixing network that leverages pixel- and window-level attention to improve spatial feature extraction and HU task. Tao et al. [42] enhanced HU accuracy through a spectral–spatial attention network. Dong et al. [43] improved HU performance using a graph-based adaptive attention model. Benhalouche et al. [44] developed an new ADMM-based unmixing algorithm using a multiplicative LSMM to handle spectral variability. Xu and Na [45] proposed a convolution-Transformer-based reconstruction and sparse unmixing method to handle spectral variability and spatial heterogeneity of end-members. To enhance interpretability, Xiong et al. [46] introduced algorithm unrolling, where the parameters of submodules are alternately optimized, thereby improving the accuracy of abundance estimation. Subsequently, Shu et al. [4] proposed a Nesterov-accelerated unrolling network that reduces computational complexity and accelerates convergence. Notably, these methods estimate end-members and abundances based on weights/outputs from specific layers, typically initializing decoder weights (i.e., end-members) using vertex component analysis (VCA) [47]. Consequently, their performance heavily relies on accurate initial end-members [48]; otherwise, abundance estimation degrades substantially. Moreover, estimating abundances solely through decoder weights overlooks spatial distribution patterns of land cover [9], [35].

Breakthroughs in large-scale DL methods have profoundly transformed natural image understanding, with representative examples such as masked autoencoders (MAEs) [49] and the segment anything model (SAM) [50]. Inspired by their success, the remote sensing community has begun to adapt these foundational models to a variety of downstream tasks [51], [52]. For instance, Ma et al. [53] introduced the SAM-driven extraction mechanism into hyperspectral few-shot classification, which reduces label dependency while improving classification accuracy. Zhu et al. [51] employed SAM to capture global spatial features of land cover distributions in HSIs. Wang et al. [54] applied SAM for pixel-level annotation of remote sensing imagery. Ma et al. [53] exploited SAM to promote interactions between spatial locations and semantic features. In addition, Zou et al. [55] developed a semiautomatic annotation strategy for remote sensing images by readapting SAM's hyperparameters. Collectively, these successful applications highlight the versatility of foundation models in remote sensing and provide valuable insights as well as potential paradigms for advancing HU research.

In advancing the interpretability of DL-based methods, many studies have proposed various innovative models or modules [56], [57]. Among these, the Kolmogorov–Arnold network (KAN) is architecturally grounded in the Kolmogorov–Arnold theorem [58], which mathematically proves that any multivariate continuous function can be represented through finite combinations of univariate continuous functions and additive operations [59]. This theoretical foundation endows KAN with dual advantages: it effectively captures complex nonlinear relationships between input and target data while maintaining parameter efficiency through a shallow architecture [60]. These distinctive architectural characteristics make KAN particularly suitable for handling the inherent nonlinear complexities in HSI processing tasks, eliminating the need for deep architectures and their associated computational overhead [61], [62].

Current DL-based blind HU methods heavily rely on well-initialized end-members, and their performance shows significant uncertainty when such initializations, like VCA, are absent [48], [63]. Considering the strengths of KAN and foundational large models, this article proposes a blind HU method that integrates Recurrent-KAN with an SAM adapter. Adopting the LSMM, the proposed method employs Recurrent-KAN to decompose spectral features from mixed pixels and utilizes SAM encoder to capture spatial distribution feature of land cover, thereby enabling the transformation from "spectral features to end-member extraction" and "spatial features to abundance estimation." The innovations are summarized as follows.

1) A novel HU framework is proposed, which enables the joint inference of latent end-members and their abundances in a fully data-driven manner. Requiring only the number of end-members as input, spectral signatures and their spatial distributions are autonomously learned.
2) A Recurrent-KAN module is designed, in which high-dimensional functional decomposition of mixed pixels is performed and outputs from the previous stage are integrated to prevent short-term forgetting, enabling adaptive reconstruction of target end-members with improved stability.
3) An HU-SAM adapter is developed for spatial feature extraction, which achieves global spatial awareness while capturing gradient-level details to restore fine-grained textures, thereby enabling more precise delineation of land-cover spatial distributions.

The remainder of this article is organized as follows. Section II introduces the preparatory knowledge. Section III details the proposed method. Section IV describes the experimental settings and comparison. Section V discusses and analyzes the results. Finally, Section VI concludes the work.

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II. Preparatory Knowledge

A. Kolmogorov–Arnold Representation Theorem

The Kolmogorov–Arnold theorem, also known as the superposition theorem, is a fundamental result in approximation theory. It asserts that any multivariate continuous function on a bounded domain can be represented as a finite composition of univariate functions and linear operations (additions) [62]. Specifically, for any smooth function $f : [0,1]^n \leftarrow \mathbb{R}$, it can be expressed as follows:

$$f(x) = \sum_{q=1}^{2n_{\text{in}}+1} \psi_q \left(\sum_{p=1}^{n_{\text{in}}} \phi_{p,q}(x_p) \right) \tag{1}$$

where $x = x_1, x_2, \ldots, x_{d_{\text{in}}}, \phi_{p,q} : [0,1] \rightarrow \mathbb{R}$, and $\psi_q : \mathbb{R} \rightarrow \mathbb{R}$. The condition for the realization of this theorem is that these univariate functions are continuous functions.

B. Kolmogorov–Arnold Network

Although the Kolmogorov–Arnold theorem indicates that any continuous multivariate function on a bounded domain can be decomposed into a superposition and addition of univariate continuous functions [64], their specific forms are problem-dependent and thus not directly applicable in practice [65].

Liu et al. [65] operationalized and extended the Kolmogorov–Arnold theorem within deep networks. The resulting KAN incorporates spline functions and basis functions with learnable coefficients at each layer, expressed as follows:

$$\phi(y) = w_b b(y) + w_s \text{spline}(y) \tag{2}$$

where w_b and w_s are learnable weights controlling the overall amplitude of activation function ϕ. The basis function $b(y)$ employs the sigmoid linear unit (SiLU)

$$b(y) = \text{SiLU}(y) = \frac{y}{1+e^{-y}}. \tag{3}$$

The spline function is parameterized as a linear combination of B-spline basis function

$$\text{spline}(y) = \sum_i c_i B_i(y) \tag{4}$$

where $B_i(y)$ are basis functions constructing univariate continuous functions, and c_i denote corresponding weights [61]. Through learnable coefficients (w_b, w_s, c_i), the network autonomously adjusts its parameters to optimize data fitting.

Conceptually, KAN adaptively processes input signals through B-spline functions, generating output via linear combinations of linear transforms and spline transformations. Emerging research highlights KAN's advantages in tasks demanding precise function approximation [66].

In the application of remote sensing, Seydi et al. [61] demonstrated that KAN can enhance the accuracy of hyperspectral change detection tasks by effectively capturing the nonlinear separability between changed and unchanged classes, while maintaining computational efficiency. Zhao et al. [67] exploited the smooth nonlinear characteristics of KAN to suppress gradient perturbations and to capture spatial–spectral features, thereby significantly improving the classification robustness and accuracy of HSIs under strong interference conditions. Wu et al. [68] integrated KAN with CNN within an asymmetric convolutional autoencoder framework, effectively enhancing background reconstruction and anomaly detection performance in HSIs. These studies collectively demonstrate the promising potential of KAN in various remote sensing tasks.

C. Linear Spectral Mixture Model

In the LSMM, macroscopic pure components are assumed to be uniformly distributed within individual pixels in the field of view. Each photon interacts with only one pure material before reaching the observer, unaffected by any other substances. The LSMM is expressed as follows:

$$\mathbf{y} = a_1\mathbf{m_1} + a_2\mathbf{m_2} + \ldots + a_n\mathbf{m_n} = \mathbf{Ma} \tag{5}$$

where $\mathbf{M} = [\mathbf{m_1}, \ldots, \mathbf{m_n}]$ is the end-member matrix containing n end-members, and $\mathbf{a} = [a_1, \ldots, a_n]^T$ denotes the abundance vector. Thus, the observed spectral signature $\mathbf{y}$ of a pixel is a linear combination of pure materials $\mathbf{m_i}$ weighted by their abundances a_i. The generalized form accounting for pixel-varying end-members is given by

$$\mathbf{Y} = \mathbf{MA} \tag{6}$$

where $\mathbf{Y}$ represents the HSI, and $\mathbf{A} = [\mathbf{a}_1, \ldots, \mathbf{a}_N](1 \le i \le N)$ is the abundance matrix for N pixels.

Without band removal, each end-member $\mathbf{m}$ represents a continuous spectral reflectance signal captured across contiguous wavelength bands. This signal manifests as a continuous and smooth curve in the spectral dimension [69]. Furthermore, the distinct spectral characteristics exhibited by different materials enable their identification and discrimination, providing the foundation for feature extraction along the spectral dimension.

D. SAM Image Encoder

With substantial expansion of data and model scales, foundational models have achieved remarkable success across diverse domains, including computer vision. Pretrained on large-scale datasets, these models typically adapt to multiple applications through parameter fine-tuning or specialized "adapters," demonstrating performance comparable to or surpassing task-specific models [70]. The SAM, trained on 11 million images with over one billion masks, delivers powerful visual feature understanding and segmentation capabilities [71]. Its architecture comprises three core components: 1) a powerful image encoder [MAE-pretrained Vision Transformer (ViT)] for extracting hierarchical features; 2) a prompt encoder integrating positional and mask embeddings; and 3) a mask decoder implemented with a bidirectional Transformer.

The core design of SAM image encoder lies in the ViT, which partitions images into $P \times P$ patches, flattens them, and maps to a high-dimensional space via linear projection. ViT's efficacy stems from its multihead self-attention (MSA) and feed-forward network (FFN). The MSA mechanism captures global dependencies among patches

$$\text{MSA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)\mathbf{W_O} \tag{7}$$

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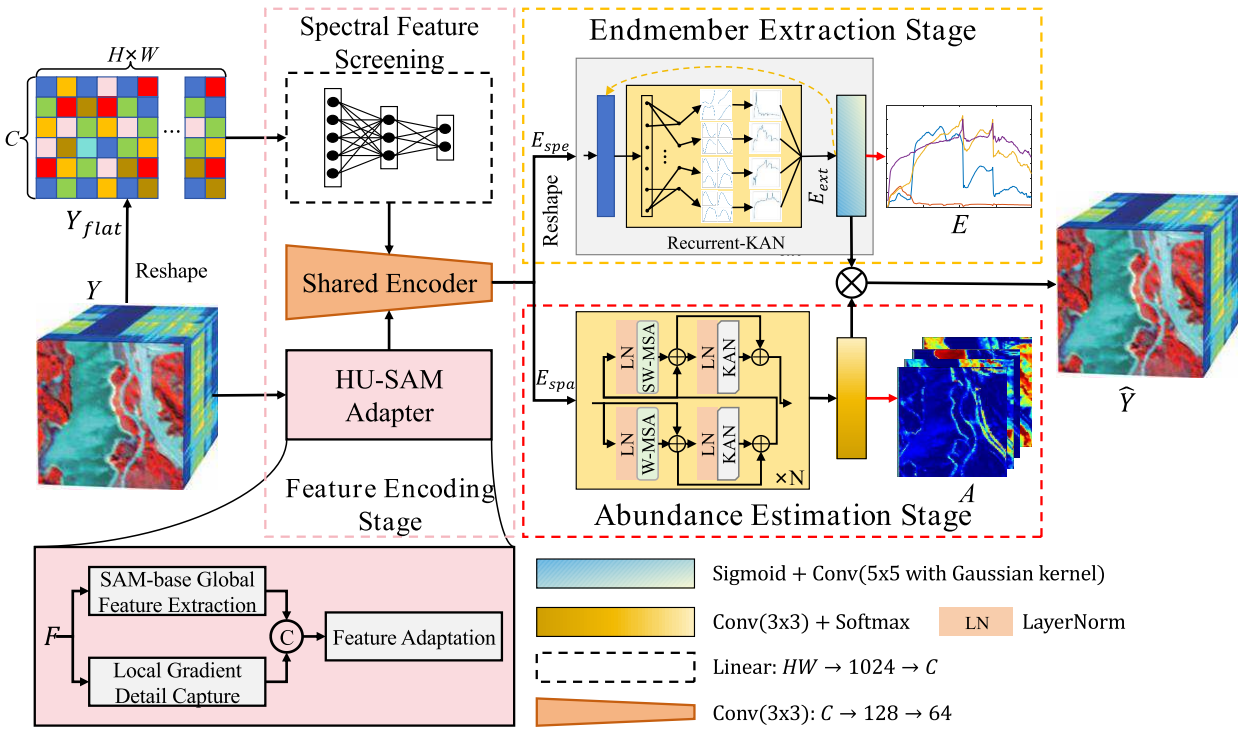


Fig. 1. Overall structure of the proposed method. The network consists of three components: the feature encoding stage for extracting spatial features E_{spa} and spectral features E_{spe}, the abundance estimation stage for generating the abundance map $\mathbf{A}$, and the end-member extraction stage for obtaining end-members $\mathbf{E}$. Finally, the reconstructed image $\hat{\mathbf{Y}}$ is produced based on the LSMM.

where $\mathbf{Q}, \mathbf{K}$, and $\mathbf{V}$ are query, key, and value matrices for computing attention weights, and $\mathbf{W_O}$ is the output projection matrix. Each head_i computes patchwise similarities

$$\text{head}_i = \text{Attention}\left(\mathbf{QW_Q^i}, \mathbf{KW_K^i}, \mathbf{VW_V^i}\right) \tag{8}$$

where $\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}((\mathbf{QK^T}/(d_k)^{1/2}))\mathbf{V}$, $\mathbf{W_Q^i}, \mathbf{W_K^i}, \mathbf{W_V^i} \in R^{D\times d_k}$ are projection matrices for the head_i, $d_k = D/k$, and h is the number of heads.

The FFN applies nonlinear transformations to each embedding

$$\text{FFN}(x) = \max(0, \mathbf{W_1}x + b_1)\,\mathbf{W_2} + b_2 \tag{9}$$

where $\mathbf{W_1}, \mathbf{W_2}$ are weight matrices and b_1, b_2 are bias terms.

Pretrained with MAE, SAM encoder learns structural continuity (e.g., contours and textures) by reconstructing randomly masked patches. This equips SAM with few-shot and zero-shot generalization [70], enabling interpretation of unseen objects through learned structural priors.

In hyperspectral remote sensing, spatial continuity—exemplified by road networks with defined orientations and buildings with clustered geometries—is a fundamental characteristic. Leveraging SAM's pretrained encoder allows efficient extraction of spatial features, facilitating precise separation of material compositions through its acquired perception of structural continuity.

III. Proposed Method

This section details the proposed framework, including its design motivation and architectural specifics. As illustrated in Fig. 1, the network operates through three stages: feature encoding, end-member extraction, and abundance estimation.

The feature encoding stage aims to preliminarily perform spectral feature screening (SFS) on the pixel-composed matrix and spatial feature extraction on the band-composed HSI, respectively, followed by a shared encoder generating spectral features E_{spe} and spatial features E_{spa}; subsequently, the end-member extraction stage decomposes and reconstructs E_{spe} to output smooth end-member signatures $\mathbf{E}$; furthermore, the abundance estimation stage processes E_{spa} through global–local spatial feature extraction to derive abundance estimates $\mathbf{A}$; finally, based on the LSMM, the reconstruction $\hat{\mathbf{Y}} = \mathbf{A} \times \mathbf{E}$ is optimized via the loss function L_m.

A. Motivation and Hypothesis

The abundance of mixed pixels in hyperspectral data provides a data foundation for neural networks to capture and separate spectral features, while foundation models' powerful understanding of object characteristics in images has been widely demonstrated. Leveraging the functional decomposition capability of the Kolmogorov–Arnold theorem and SAM's capacity to capture spatially continuous features, we hope that the DL-based model can spontaneously achieve the dual tasks of end-member extraction and abundance estimation, ultimately attaining the conversion objective of transforming spectral features into end-member signatures and spatial features into abundance results.

Integrating Kolmogorov–Arnold theorem with hyperspectral data acquisition properties, we establish the following.

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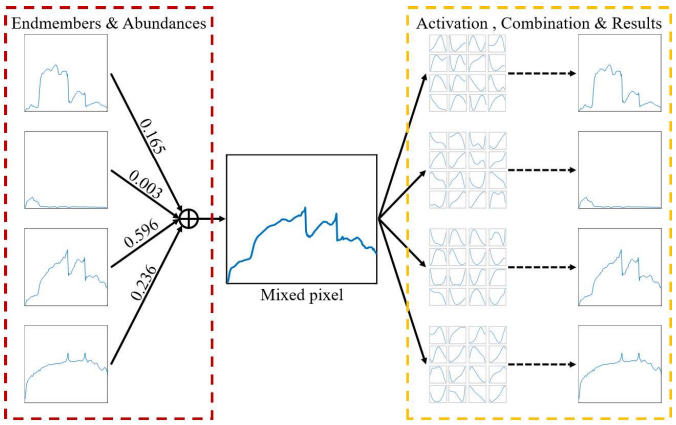


Fig. 2. Schematic of the proposed method. The left part illustrates the process of generating mixed pixels from pure end-members based on the LSMM. The right part shows the process of extracting and combining pure end-members from the spectra of mixed pixels using univariate functions.

Assumption 1: Any end-member can be represented as a linear superposition of finite distinct univariate functions

$$\mathbf{m} = \sum_q \psi \left(\sum_p \lambda_q \phi \left(x_p + q \cdot \triangle \right) \right) \tag{10}$$

where

1) $\psi(\cdot)$: Outer function (spectral composition function);
2) $\phi(\cdot)$: Inner function (elementary feature function);
3) $\phi(x_p + q \cdot \triangle)$: Univariate function (characterizing local spectral features);
4) $\lambda_q, \triangle \in \mathbb{R}$: Learnable parameters;
5) x_p: Spectral sampling point.

Combining (5) and Assumption 1 yields

$$\begin{aligned} \mathbf{y} &= \mathbf{Ma} \\ &= \left[\sum_q \psi_1 \left(\sum_p \lambda_q \phi_1 \left(x_p + q_1 \cdot \triangle_1 \right) \right), \right. \\ &\quad \left. \ldots, \sum_q \psi_n \left(\sum_p \lambda_q \phi_n \left(x_p + q_n \cdot \triangle_n \right) \right) \right] \cdot [a_1, \ldots, a_N]. \end{aligned} \tag{11}$$

This demonstrates that a hyperspectral mixed pixel can be linearly decomposed by spectral features represented through finite univariate functions, which can reconstruct end-members via specific linear superpositions, providing a potential possibility for applying KAN to HU task, as shown in Fig. 2.

B. HU-SAM Adapter

The proposed HU-SAM adapter, illustrated in Fig. 3, comprises three components: SAM-based global feature extraction (SAM-BGFE), local gradient detail capture (LGDC), and feature adaptation (FA). Given input inner features $F_{\text{in}_i} \in \mathbb{R}^{C \times H \times W}$ from the ith layer, it outputs $F_{\text{in}_{i+1}} \in \mathbb{R}^{C \times H \times W}$.

1) SAM-Based Global Feature Extraction: First, gradient information Grad (computed via Sobel operator from Y) enhances spatial details

$$F'_{\text{in}_i} = F_{\text{in}_i} \times (1 + \text{Grad}). \tag{12}$$

Next, channelwise feature selection (FS) upsamples F'_{in_i} to $[1024, 1024, 3]$ for SAM input. After SAM encoding (weights frozen), downsampling restores dimensions, followed by a projection layer

$$F_g = \text{proj}_{\cdot} \left(\downarrow \text{Encoder}_{\text{SAM}} \left(\uparrow \text{FS} \left(F'_{\text{in}_i} \right) \right) \right) \tag{13}$$

where $\downarrow$ and $\uparrow$ denote bicubic interpolation, ensuring stable feature representation.

2) Local Gradient Detail Capture: This component, which introduces gradient centralized convolution (GCConv) [72], is enhanced by dual spatial attention [73]. It aims to further improve the local texture extraction capability of spatial attention, thereby mitigating SAM's limitation in capturing sparse objects (e.g., trees on soil).

For input feature $X \in \mathbb{R}^{C_{\text{in}} \times H \times W}$ and kernel $K \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times H \times W}$, its output feature is $Y \in \mathbb{R}^{C_{\text{out}} \times H \times W}$. For the normal convolution operation, $Y_{\text{normal}} = X * K$, where $*$ means the convolution operation.

Summing the convolution kernel K over spatial dimensions yields the gradient-centralized kernel $K_{\text{diff}} \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}}}$

$$K_{\text{diff}}[C_{\text{out}}, C_{\text{in}}] = \sum_{i=0}^{k-1} \sum_{j=0}^{k-1} K\left[C_{\text{out}}, C_{\text{in}}, i, j\right]. \tag{14}$$

The get the output $Y_{\text{diff}} = X * K_{\text{diff}}$. And, the final output $Y = Y_{\text{normal}} - \theta Y_{\text{diff}}$, where θ controls the influence strength of gradient centralization.

The LGDC module is formulated as follows:

$$F_l = F_{\text{in}_i} \times \sigma \left(W_{\triangle} \left(\text{Max}1d \left(F_{\text{in}_i} \right), \text{Avg}1d \left(F_{\text{in}_i} \right) \right) \right) \tag{15}$$

where $W_{\triangle}$ means GCConv, $\text{Max}1d(F_{\text{in}_i}), \text{Avg}1d(F_{\text{in}_i})$ mean channelwise maximum/average pooling, and σ means Sigmoid activation.

3) Feature Adaptation: Concatenated features $[F_g, F_l]$ undergo scale-invariant processing

$$F_f = W_1 \left(\text{concat} \left(F_g, F_l \right) \right) \tag{16}$$

$$F_{\text{up}} = \text{PSup} \left(\text{Max}2d \left(F_f \right), \text{Avg}2d \left(F_f \right) \right) \tag{17}$$

$$F_{\text{in}_{i+1}} = \text{GeLU} \left(\text{BN} \left(F_{\text{up}} + F_f \right) \right) + F_{\text{in}_i} \tag{18}$$

where $\text{PSup}(\cdot)$ is the pixel shuffle upsampling, and W_1 denotes 1×1 convolution.

C. Feature Encoding Stage

This stage employs the HU-SAM adapter to capture spatial features F_{spa}

$$F_{\text{spa}} = \text{Adapter}(\mathbf{Y}, \text{Grad}) \tag{19}$$

where $\mathbf{Y}, F_{\text{spa}} \in \mathbb{R}^{C \times H \times W}$.

Simultaneously, spectral features F_{spe} are extracted via

$$F_{\text{spe}} = \text{SFS}(\mathbf{Y}_{\text{flat}}) \tag{20}$$

with $\mathbf{Y}_{\text{flat}} \in \mathbb{R}^{C \times \text{HW}}$ obtained through identical transformation of $\mathbf{Y}$. The SFS compresses the pixel matrix along the spatial dimension using two linear layers, yielding $\dim(F_{\text{spe}}) = [C, C]$ (first C: number of pixels, second C: number of bands).

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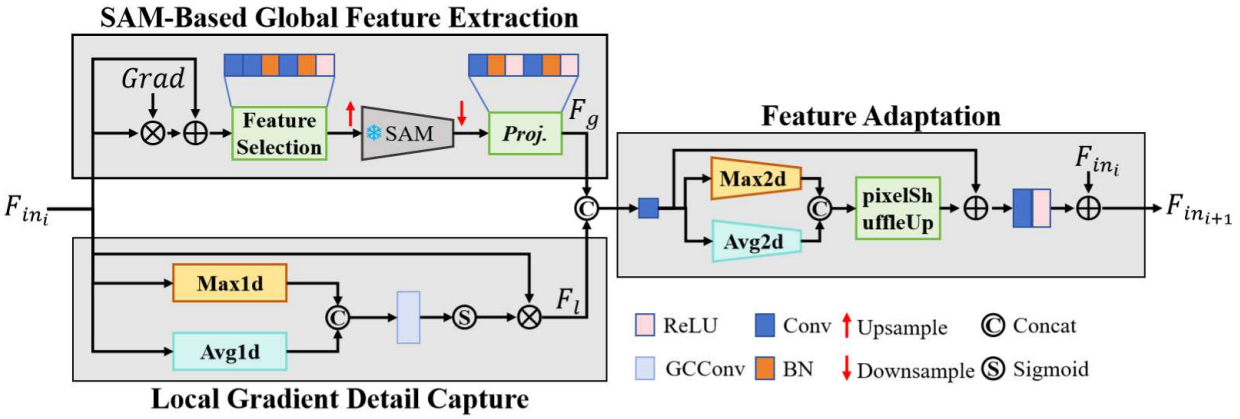


Fig. 3. Design of HU-SAM adapter.

To enable feature interaction between F_{spe} and F_{spa}, we perform dimension expansion $F_{\text{spe}} \xrightarrow{\text{expand}} \mathbb{R}^{C\times C\times 1\times 1}$ (interpretation: [num, bands, 1, 1]) and $F_{\text{spa}} \xrightarrow{\text{expand}} \mathbb{R}^{1\times C\times H\times W}$ (interpretation: [batch, bands, height, width]).

The shared encoder S-encoder uses 1×1 convolutions for cross-dimensional processing

$$E_{\text{spe}} = \text{S-encoder}\left(F_{\text{spe}}\right) \tag{21}$$
$$E_{\text{spa}} = \text{S-encoder}\left(F_{\text{spa}}\right) \tag{22}$$

where $\text{S-encoder}(x) = \text{LReLU}(W_1(\text{LReLU}(\text{BN}(W_x(x)))))$.

D. End-Member Extraction Stage

In this stage, we utilize the Recurrent-KAN to extract end-members.

As recent research [66] indicates, while KAN demonstrates superior learning efficiency, it exhibits significant memory forgetting. To address this limitation, we propose Recurrent-KAN (Fig. 1), which concatenates E_{ext}^{i-1} from the previous iteration with current spectral features E_{spe} through concatenation, thereby enhancing stability in end-member estimation. It is defined as follows:

$$E_{\text{ext}}^{i} = \text{R-KAN}\left(W_1\left(\text{Concat}\left(E_{\text{ext}}^{i-1}, E_{\text{spe}}\right)\right)\right) \tag{23}$$
$$\mathbf{E} = W_{\text{GS}}\left(\sigma\left(E_{\text{ext}}^{i}\right)\right) \tag{24}$$

where R-KAN means Recurrent-KAN, which is set to $[32, 64, p]$, following the layer specifications in (2). Here, p represents the count of end-members. E_{ext}^{0} is initialized randomly. Additionally, W_{GS} performs a 5×5 Gaussian kernel convolution to refine $\mathbf{E}$ through smoothing.

E. Abundance Estimation Stage

This stage leverages E_i within the LSMM framework to transform spatial features E_{spa} into abundance maps aligned with $\mathbf{E}$. The network architecture adapts the Swin Transformer structure from SwinT [38], where the Window MSA (W-MSA) extracts localized spatial patterns by partitioning features into nonoverlapping $k\times k$ patches for attention computation, while the Shifted Window MSA (SW-MSA) enhances global context awareness through $\lfloor M/2, M/2\rfloor$-pixel window shifting to facilitate cross-patch interactions.

To optimize parameter efficiency, we substitute the original MLP layers in Swin Transformer with KAN blocks following [65]. It can be expressed as follows:

$$\hat{z}^{l} = \text{W-MSA}\left(LN\left(z^{l-1}\right)\right) + z^{l-1} \tag{25}$$
$$z^{l} = \text{KAN}\left(LN\left(\hat{z}^{l}\right)\right) + \hat{z}^{l} \tag{26}$$
$$\hat{z}^{l+1} = \text{SW-MSA}\left(LN\left(z^{l}\right)\right) + z^{l} \tag{27}$$
$$z^{l+1} = \text{KAN}\left(LN\left(\hat{z}^{l+1}\right)\right) + \hat{z}^{l+1}. \tag{28}$$

A smoothing layer integrates a 3×3 convolution with Softmax activation to enforce the "sum-to-one" constraint on abundance results $\mathbf{A}$. Under the LSMM, the estimated HSI $\hat{\mathbf{Y}}$ can be reconstructed as $\hat{\mathbf{Y}} = \mathbf{A}\times\mathbf{E}$.

F. Loss Function Design

The proposed method integrates four complementary loss functions: 1) reconstruction loss to ensure accurate spectral reconstruction; 2) smoothing regularization to enhance sensitivity to material boundaries and structural information [74]; 3) sparsity constraint to improve model stability and robustness; and 4) end-member decorrelation to prevent uneven distribution of extracted end-members

$$\begin{aligned}\mathcal{L}_{\text{m}}\left(\mathbf{Y},\hat{\mathbf{Y}},\mathbf{A},\mathbf{E}\right) = &\underbrace{\text{SADLoss}(\mathbf{Y},\hat{\mathbf{Y}}) + \text{MSE}(\mathbf{Y},\hat{\mathbf{Y}})}_{\text{Reconstruction Loss}} \\ &+ \alpha\cdot\underbrace{\sum_{x,y}\sqrt{\left(\frac{\partial\mathbf{A}}{\partial x}\right)^2+\left(\frac{\partial\mathbf{A}}{\partial y}\right)^2}}_{\text{Total Variation (TV) Loss}} \\ &+ \beta\cdot\underbrace{\sum_{i=1}^{P}\sqrt{\|\mathbf{A}_i\|_1}}_{\text{Sparsity Regularization}} \\ &+ \gamma\cdot\underbrace{\frac{1}{p(p-1)}\sum_{i=1}^{p}\sum_{j=i+1}^{p}D_{\text{KL}}(\mathbf{e}_i\,\|\,\mathbf{e}_j)}_{\text{Endmember Decorrelation}}\end{aligned} \tag{29}$$

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where $D_{\mathrm{KL}}(\mathbf{e}_i \| \mathbf{e}_j) = \sum_{k=1}^{c} \mathbf{E}(k) \log(\mathbf{e}_i(k)/\mathbf{e}_j(k))$, and $\mathbf{e}_i(k)$ represents the value of end-member $\mathbf{E}$ at the kth band of the ith spectrum.

Hyperparameters are set as $\alpha = 10^{-8}, \beta = 10^{-7}$, and $\gamma = 10^{2}$.

IV. Experimental Settings and Comparison

A. Dataset Settings

Three datasets are used to evaluate the proposed method, namely Jasper Ridge, Urban, and Synthesis.

1) *Jasper Ridge Dataset [75]:* The data was collected over Central California, USA, using the AVIRIS sensor. The scene comprises four materials: water, soil, trees, and road. This HSI consists of 100 × 100 pixels, with the original data comprising 224 bands. The spectral range spans from 0.38 to 2.5 μm. After removing the effects of water vapor and atmospheric interference, the data retains 198 bands.
2) *Urban Dataset [76], [77]:* The data was collected over the urban area of Copperas Cove, Texas, USA, using the HYDICE HSI sensor. The scene comprises four materials: asphalt, grass, trees, and roof. This HSI consists of 307 × 307 pixels. After removing dense water vapor and atmospheric effects, the dataset retains 162 bands, with a spectral range of 400 ~ 2500 nm and a spectral resolution of 10 nm.
3) *Synthesis Dataset:* This dataset is a synthetic one, containing a variety of mineral standard end-member curves from the United States Geological Survey spectral library. The dataset can be utilized with its built-in multilinear mixing model to generate datasets of different sizes. In this proposed method, we selected "Spheric Gaussian Field" dataset provided by this data source for experiments. The dataset simulates realistic hyperspectral scenes through linear mixing of five distinct end-members: asphalt, fiberglass, plastic, sheet metal, and brick. The acquired HSI comprises a 128 × 128 pixel grid spanning 431 spectral bands (wavelength range: 0.4 ~ 2.5 μm).

B. Quantitative Metrics

We adopt two metrics—spectral angle distance (SAD) and root mean square error (RMSE)—for objective performance comparison. The mean metrics with standard deviations (mSAD±STD and mRMSE±STD) summarize results across all land covers, with ideal values theoretically approaching zero.

SAD can quantify spectral fidelity between estimated end-members $\hat{\mathbf{E}}$ and standard end-member $\mathbf{E}$

$$\mathrm{SAD}\left(\hat{\mathbf{E}}, \mathbf{E}\right) = \frac{1}{p} \sum_{i=1}^{p} \arccos\left(\frac{\langle \mathbf{E_i}, \hat{\mathbf{E}}_\mathbf{i} \rangle}{\|\mathbf{E_i}\|_2, \|\hat{\mathbf{E}}_\mathbf{i}\|_2}\right) \tag{30}$$

where $\hat{\mathbf{E}}_\mathbf{i}$ and $\mathbf{E_i}$ denote the ith estimated and reference end-member, respectively, and p represents end-member count.

The RMSE evaluates spatial accuracy of abundance maps $\hat{\mathbf{A}}_\mathbf{i}$ against true $\mathbf{A_i}$

$$\mathrm{RMSE}\left(\hat{\mathbf{A}}, \mathbf{A}\right) = \frac{1}{p} \sum_{i=1}^{p} \sqrt{\left\|\mathbf{A_i} - \hat{\mathbf{A}}_\mathbf{i}\right\|^2} \tag{31}$$

where $\hat{\mathbf{A}}_\mathbf{i}$ and $\mathbf{A_i}$ represent the ith estimated and ground-truth abundance results.

C. Comparative Methods and Parameter Settings

We compare with eight methods, including PLMM [78], 3DCNN [37], SSABN [39], SwinT [38], A2SAN [48], DFFN [79], SSAF [80], and MGANet [40]. Notably, A2SAN, DFFN, SSAF, and our proposed method do not require end-member initialization, whereas the remaining methods do. To ensure fair comparison, VCA initialization is applied to methods that require end-member initialization. All methods adopt default parameter configurations.

For Jasper Ridge dataset, our method trains for 350 epochs with a global learning rate of 0.002 and HU-SAM adapter learning rate of 0.004. For Urban dataset, training lasts 500 epochs with global learning rate of 0.004 and adapter learning rate of 0.006. For Synthesis dataset, 800 epochs are conducted with global learning rate of 0.005 and adapter learning rate of 0.025. All methods employ cosine annealing scheduler and AdamW optimizer, with learning rate decay cycles equal to iteration count × 10. The learning process is terminated once the reconstruction loss converges (loss change < 10^{-5}) or the maximum number of epochs is reached.

D. Experiments on the Jasper Ridge Dataset

Fig. 4 displays the performance of all methods on the Jasper Ridge dataset. In terms of abundance results, PLMM fails to estimate "road" in its results. 3D-CNN exhibits significant overestimation of "road," SSABN fails to estimate the "soil," and both SwinT and A2SAN struggle to separate "road" from "soil" in their abundance results, which leads to misclassification of some "road" pixels within the "soil." DFFN's estimation of "road" exhibits inaccuracies. MAGNet's abundance result exists in erroneous. Although the proposed method did not achieve optimal/suboptimal performance in "water" abundance estimation, visual interpretation reveals that the estimation errors are primarily concentrated in "road" with misclassifications, while the abundance estimation of "water" remains fundamentally accurate. Considering the overall mapping performance, the proposed method demonstrates superior effectiveness compared to existing comparative approaches.

Fig. 5 shows the end-members obtained by different methods in various ground features. Table I quantitatively compares the performance metrics of all methods on the Jasper Ridge dataset. The quantitative analysis demonstrates that most methods show elevated SAD values and abundance errors in "road" (excluding 3D-CNN, SSAF, and proposed method), while our method outperforms all competitors across comprehensive evaluation metrics.

E. Experiments on the Urban Dataset

Figs. 6 and 7 visually compare the performance of all methods on the Urban dataset.

The ground feature distribution on Urban dataset exhibits greater complexity than that on Jasper Ridge dataset. PLMM

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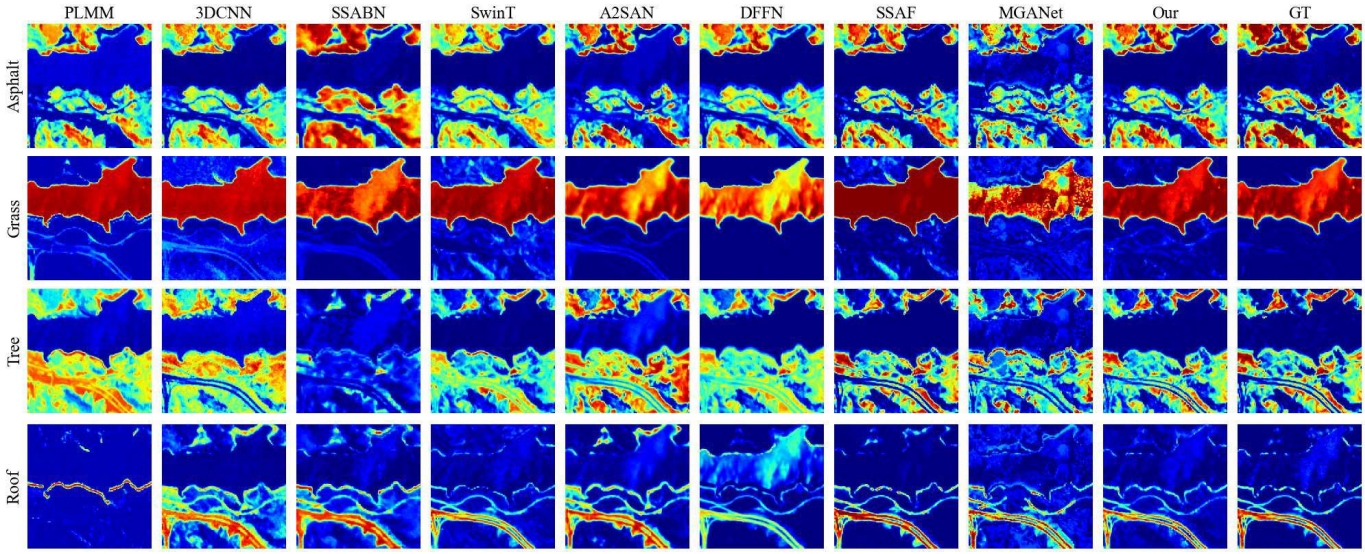


Fig. 4. Abundance estimation results of different methods on the Jasper Ridge dataset.

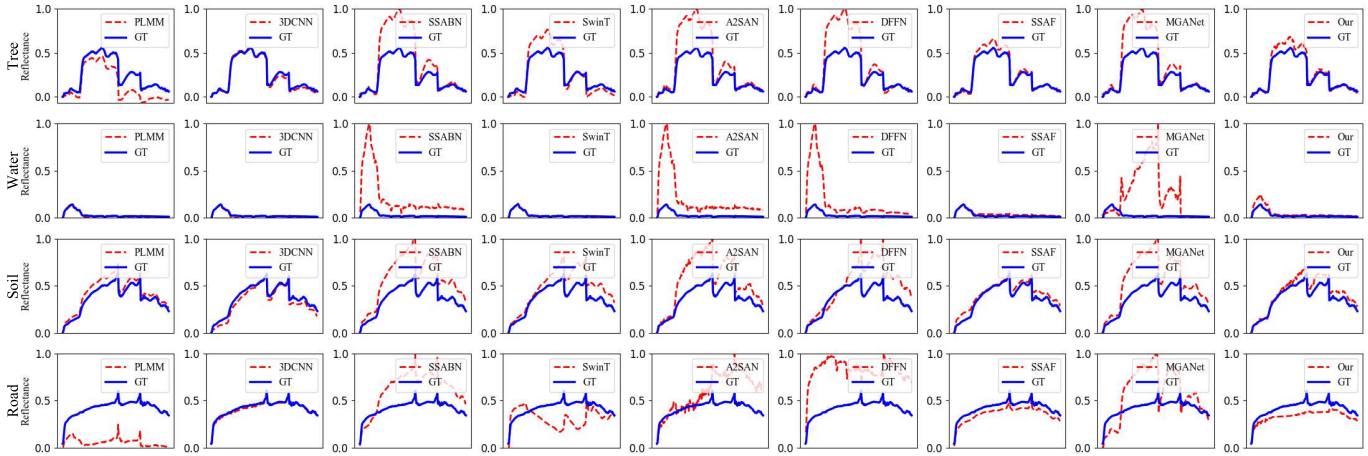


Fig. 5. End-members obtained from various ground features by different methods on the Jasper Ridge dataset. The *X*-axis represents the wavelength range.

TABLE I

COMPARISON OF DIFFERENT METHODS ON THE JASPER RIDGE DATASET USING SAD/RMSE

Method	Tree	Water	Soil	Road	mSAD±STD	mRMSE±STD
PLMM	0.390/0.201	0.360/0.117	0.036/0.313	0.586/0.266	0.343±0.228	0.224±0.085
3DCNN	0.072/0.165	0.434/0.126	0.073/0.221	**0.025**/0.173	0.151±0.190	0.171±0.039
SSABN	0.100/0.132	0.057/**0.051**	**0.042**/0.254	0.152/0.149	0.088±0.049	0.147±0.083
SwinT	0.157/0.132	0.099/0.124	0.092/0.137	0.359/0.116	0.177±0.125	0.127±0.009
A2SAN	0.103/0.137	0.077/0.066	0.104/0.154	0.224/0.114	0.127±0.066	0.118±0.038
DFFN	0.141/0.094	0.154/0.098	0.210/0.122	0.156/0.131	0.165±0.031	0.111±0.018
SSAF	**0.049**/0.105	0.239/0.115	0.085/0.085	0.030/**0.069**	0.101±0.095	0.094±0.021
MGANet	0.120/0.197	1.211/0.164	0.103/0.156	0.303/0.134	0.434±0.526	0.163±0.026
Our	0.069/**0.045**	**0.041**/0.083	0.048/**0.080**	0.029/0.070	**0.047±0.017**	**0.070±0.017**

*__Bold__ indicates the best result; underline indicates the second best.

not only exhibits low precision in abundance estimation but also suffers from various misestimation issues. For instance, PLMM incorrectly classifies "asphalt" regions as "grass" in its abundance maps. 3D-CNN shows improved abundance results but still encounters misestimation challenges. SwinT's estimated abundance lacks accuracy, A2SAN underestimates "roof," and SSABN overestimates "grass." DFFN demonstrates suboptimal accuracy for "asphalt" and "roof" abundances, while SSAF and MGANet exhibit deviations in "asphalt" estimation. Additionally, SSAF fails to provide effective abundance estimates for "roof" and "grass." Compared to existing methods, our method demonstrates superior abundance results.

Based on the end-member results presented in Fig. 7, all methods achieve reasonable "tree" extraction performance, yet slight deviations occur in "asphalt " and "roof" extraction. As

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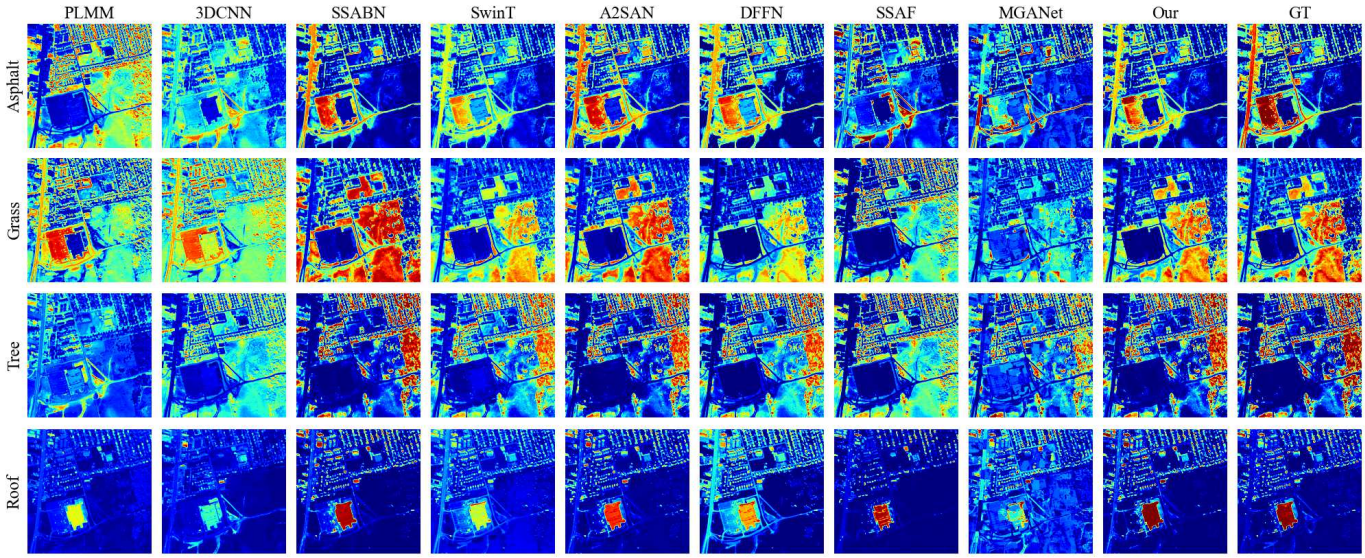


Fig. 6. Abundance estimation results of different methods on the Urban dataset.

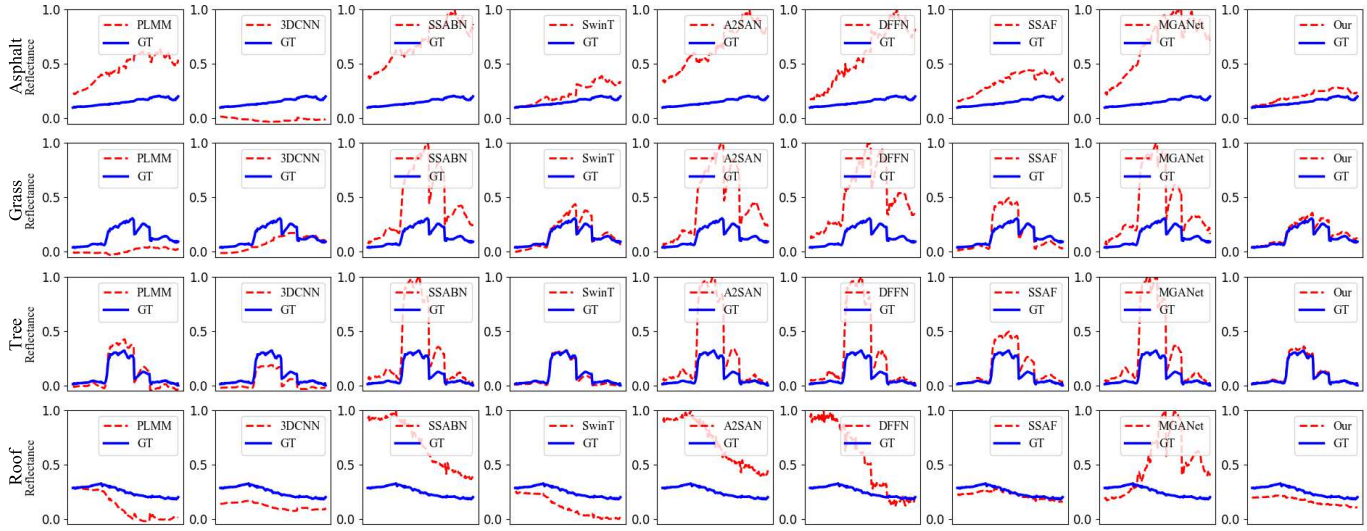


Fig. 7. End-members obtained from various ground features by different methods on the Urban dataset. The *X*-axis represents the wavelength range.

TABLE II
COMPARISON OF DIFFERENT METHODS ON THE URBAN DATASET USING SAD/RMSE

Method	Asphalt	Grass	Tree	Roof	mSAD±STD	mRMSE±STD
PLMM	0.102/0.547	1.187/0.393	0.196/0.473	0.645/0.126	0.533±0.497	0.385±0.184
3DCNN	2.415/0.274	0.497/0.393	0.359/0.245	0.096/0.183	0.842±1.062	0.274±0.088
SSABN	**0.061**/0.119	**0.057**/0.121	0.049/0.098	0.128/0.091	0.074±0.037	0.107±0.015
SwinT	0.163/0.129	0.159/0.184	0.123/0.123	0.523/0.117	0.242±0.188	0.138±0.031
A2SAN	0.086/**0.096**	0.072/0.105	0.066/0.081	0.106/**0.065**	0.083±0.018	0.087±0.018
DFFN	0.206/0.118	0.108/0.146	0.085/0.122	0.343/0.135	0.186±0.117	0.130±0.013
SSAF	0.134/0.289	0.357/0.358	0.111/0.259	0.084/0.144	0.172±0.125	0.263±0.089
MGANet	0.197/0.212	0.200/0.249	0.082/0.177	0.594/0.189	0.268±0.224	0.207±0.032
Our	0.069/0.100	0.061/**0.096**	**0.043/0.066**	**0.034**/0.075	**0.052±0.016**	**0.084±0.016**

*__Bold__ indicates the best result; underline indicates the second best.

shown in Table II, our method achieves satisfactory performance in both SAD and RMSE.

F. Experiments on the Synthesis Dataset

Figs. 8 and 9 visually compare the performance of all methods on the Synthesis dataset. The visualization results demonstrate that most methods exhibit significant discrepancies in performance when applied to the synthetic dataset. PLMM produces erroneous estimations for materials other than "fiberglass," while 3D-CNN and A2SAN demonstrate inaccuracies in all abundance estimation. Although SSABN, SwinT, and DFFN achieve relatively accurate abundance

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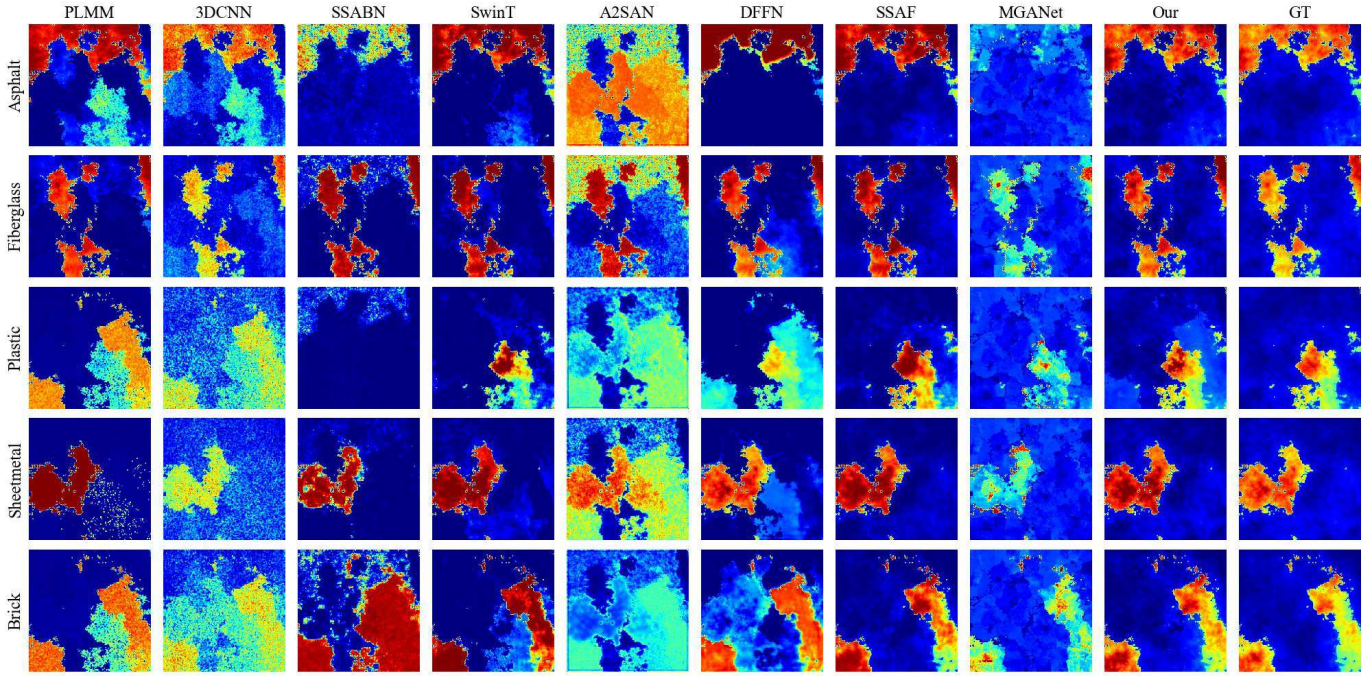


Fig. 8. Abundance estimation results of different methods on the Synthesis dataset.

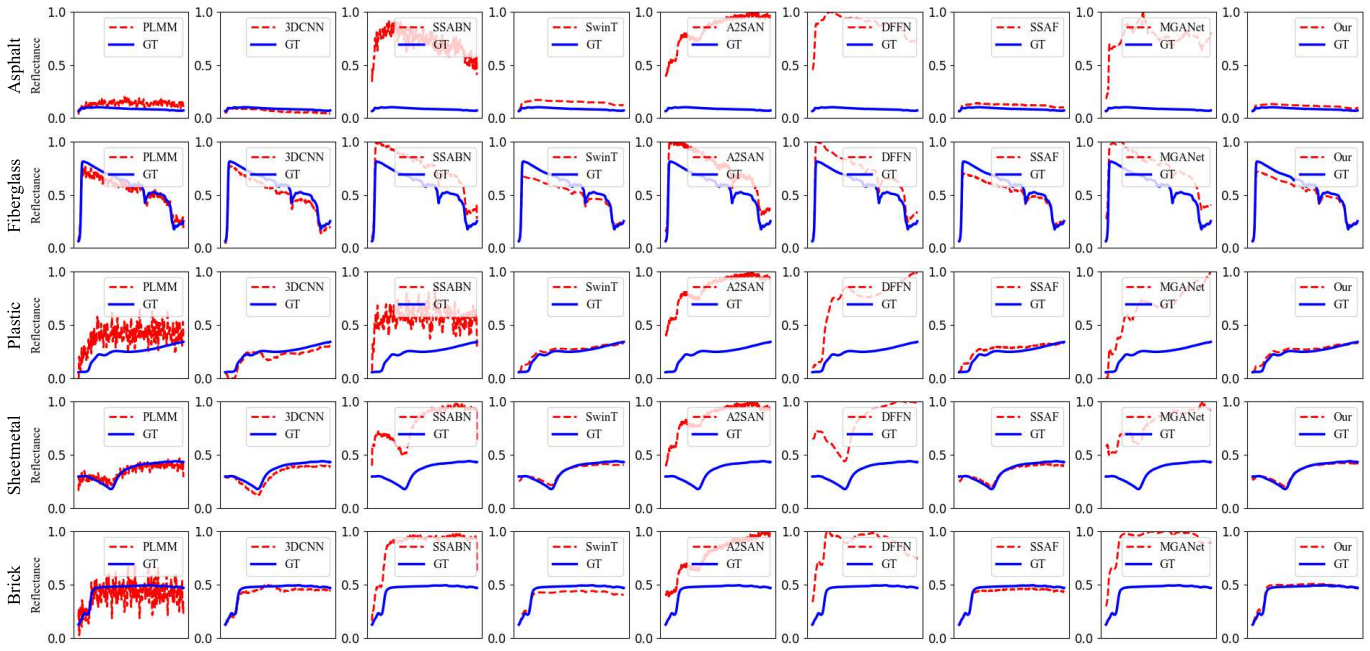


Fig. 9. End-members obtained from various ground features by different methods on the Synthesis dataset. The *X*-axis represents the wavelength range.

results for "sheet metal" and "fiberglass," substantial deviations remain in their performance across other materials. MGANet's abundance estimation accuracy requires further improvement. SSAF and the proposed method demonstrate superior abundance estimation performance compared to competing approaches.

End-member results in Fig. 9 indicate that PLMM, SSABN, A2SAN, DFFN, and MGANet all exhibit notable errors in reconstructing spectral curves for various materials. Particularly for "asphalt," "plastic," and "brick," these methods exhibit substantial errors. In Table III, the proposed method achieves satisfactory performance in both SAD and RMSE.

V. Discusses and Analyzes

Unless otherwise stated, all ablation studies follow the experimental protocol on the Jasper Ridge dataset. Hyperparameters strictly adhere to the configurations detailed in Section IV-C.

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TABLE III
COMPARISON OF DIFFERENT METHODS ON THE SYNTHESIS DATASET

Method	Tree	Water	Soil	Road	mSAD±STD	mRMSE±STD
PLMM	0.390/0.201	0.360/0.117	0.036/0.313	0.586/0.266	0.343±0.228	0.224±0.085
3DCNN	0.072/0.165	0.434/0.126	0.073/0.221	**0.025**/0.173	0.151±0.190	0.171±0.039
SSABN	0.100/0.132	0.057/**0.051**	**0.042**/0.254	0.152/0.149	0.088±0.049	0.147±0.083
SwinT	0.157/0.132	0.099/0.124	0.092/0.137	0.359/0.116	0.177±0.125	0.127±0.009
A2SAN	0.103/0.137	0.077/0.066	0.104/0.154	0.224/0.114	0.127±0.066	0.118±0.038
DFFN	0.141/0.094	0.154/0.098	0.210/0.122	0.156/0.131	0.165±0.031	0.111±0.018
SSAF	**0.049**/0.105	0.239/0.115	0.085/0.085	0.030/**0.069**	0.101±0.095	0.094±0.021
MGANet	0.120/0.197	1.211/0.164	0.103/0.156	0.303/0.134	0.434±0.526	0.163±0.026
Our	0.069/**0.045**	**0.041**/0.083	0.048/**0.080**	0.029/0.070	**0.047±0.017**	**0.070±0.017**

*__Bold__ indicates the best result; underline indicates the second best.

TABLE IV
COMPARISON OF INDIVIDUAL COMPONENTS ON THE JASPER RIDGE DATASET USING SAD/RMSE

SAM-BGFE	LGDC	FA	Tree	Water	Soil	Road	mSAD±STD	mRMSE±STD
×	×	×	**0.068**/0.068	0.060/0.092	0.071/0.091	0.043/0.083	0.061±0.013	0.084±0.011
✓	×	×	0.092/0.072	0.059/0.087	0.069/0.099	0.041/0.085	0.065±0.021	0.086±0.011
✓	✓	×	0.092/0.061	0.080/0.092	0.062/0.093	0.037/0.077	0.068±0.024	0.081±0.015
✓	✓	✓	0.069/**0.045**	**0.041/0.083**	**0.048/0.080**	**0.029/0.070**	**0.047±0.017**	**0.070±0.017**

TABLE V
COMPARISON OF DIFFERENT GRADIENT INTRODUCTION METHODS IN SAM-BGFE ON THE JASPER RIDGE DATASET USING SAD/RMSE

Input	Tree	Water	Soil	Road	mSAD±STD	mRMSE±STD
F_{in}	0.081/0.058	0.053/0.090	0.061/0.084	0.037/0.069	0.058±0.018	0.075±0.015
$F_{in} \times Grad$	0.094/0.058	0.073/0.084	0.059/0.090	0.034/0.069	0.065±0.025	0.075±0.015
$F_{in} \times (1 + Grad)$	**0.069/0.045**	**0.041/0.083**	**0.048/0.080**	**0.029/0.070**	**0.047±0.017**	**0.070±0.017**

TABLE VI
COMPARISON OF SHARED ENCODER AND GCCONV MECHANISM ON THE JASPER RIDGE DATASET USING SAD/RMSE

Shared encoder	GCConv	Tree	Water	Soil	Road	mSAD±STD	mRMSE±STD
×	✓	0.082/0.057	0.066/**0.083**	0.059/0.091	0.035/0.073	0.061±0.020	0.076±0.015
✓	×	**0.060**/0.049	0.062/0.091	0.058/0.082	0.032/0.072	0.053±0.014	0.074±0.018
✓	✓	0.069/**0.045**	**0.041/0.083**	**0.048/0.080**	**0.029/0.070**	**0.047±0.017**	**0.070±0.017**

A. Ablation Analysis of Components in HU-SAM Adaptor

We conduct ablation studies to evaluate the contributions of individual components within the HU-SAM adapter and assess critical design choices in key submodules. Table IV shows the indicators of the ablation experiment for individual components.

When SAM-BGFE is introduced, we observe slight improvements in end-member extraction and abundance results for all land covers except "tree." Subsequently, incorporating the LGDC module further reduces the RMSE metrics for "soil," "road," and "tree," indicating that the LGDC module compensates for SAM-BGFE's deficiency in capturing local information. When all modules are jointly applied, all performance metrics improve, thereby validating the rationality of the HU-SAM adapter design.

Additionally, we conduct ablation experiments to analyze the impact of incorporating gradient information and the shared encoder.

Table V illustrates how introducing gradients in SAM-BGFE affects the results. As shown in Table V, when only Y or gradient information is used as input, the proposed method fails to achieve improved accuracy. However, after enhancing edge features in Y, all evaluation metrics show improvements. This is attributed to mixed pixels commonly occurring at material boundaries—enhancing edge information thus better improves unmixing performance.

Table VI presents the results with and without the shared encoder, demonstrating that adopting the shared encoder significantly improves all evaluation metrics. For instance, in the case of "water," compared to configuration (a), the SAD improves by 37.88% while maintaining comparable abundance accuracy, and the SAD of all other land covers increases by more than 15%. Additionally, for "tree," the RMSE improves by 21.05%, and all, the mSAD and mRMSE improve by 22.73% and 8.55%, respectively. These results demonstrate that the shared encoder enhances both the end-member extraction and abundance estimation stages, further validating the inherent mapping relationship between land cover distributions and material compositions.

Table VI also presents the results of using GCConv and traditional convolution in the LDGC of the HU-SAM adapter framework. For a clearer presentation, we also conducted verification on the Urban dataset, and the results are shown in Fig. 10. On the Urban dataset with predominantly sparse object distributions, GCConv delivers substantially greater error suppression compared to the Jasper Ridge dataset featuring concentrated distributions.

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TABLE VII

COMPARISON OF USING GCCONV AND TRADITIONAL CONVOLUTION IN THE LDGC ON THE URBAN DATASET USING SAD/RMSE

	Asphalt	Grass	Tree	Roof	mSAD±STD	mRMSE±STD
Conv	**0.059**/0.230	0.065/0.119	0.052/0.075	0.169/0.240	0.086±0.055	0.166±0.082
CGConv	0.069/**0.100**	**0.061/0.096**	**0.043/0.066**	**0.034/0.075**	**0.052±0.016**	**0.084±0.016**

TABLE VIII

COMPARISON OF RECURRENT LEARNING ON THE URBAN DATASET USING SAD/RMSE

Recurrent Learning	Asphalt	Grass	Tree	Roof	mSAD±STD	mRMSE±STD
×	0.082/0.321	0.190/0.210	0.064/0.166	0.393/0.337	0.182±0.151	0.259±0.084
✓	**0.069/0.100**	**0.061/0.096**	**0.043/0.066**	**0.034/0.075**	**0.052±0.016**	**0.084±0.016**

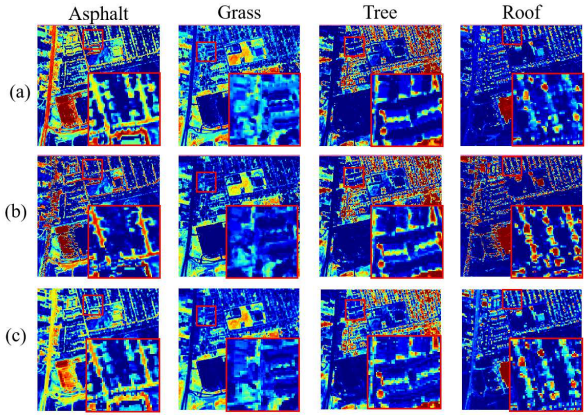


Fig. 10. Comparison of abundance results obtained by traditional convolution and GCConv on Urban dataset. (a) GT. (b) Traditional convolution. (c) GCConv. It can be seen that the abundance results using traditional convolution have obvious detail loss problems on "roof" and "asphalt," while GCConv is more obvious in capturing spatial details.

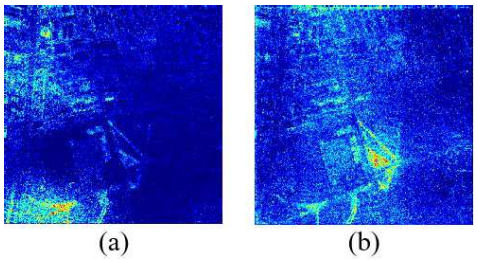


Fig. 11. Grad-CAM visualization of the HU-SAM adapter on the Urban dataset. (a) Using traditional convolution. (b) Using GCConv.

Specifically in Table VII, GCConv achieves RMSE reductions of 56.52% for "asphalt" and 68.75% for "roofs," with mean SAD decreased by 39.53% and mean RMSE by 49.39%. Table VI also exhibits significant error suppression on Jasper Ridge dataset, though less pronounced, with 11.32% SAD reduction and 5.41% RMSE reduction.

Further validation shown in Fig. 11, through Grad-CAM visualizations of the HU-SAM adapter on the Urban dataset, demonstrates GCConv's heightened sensitivity to spatially sparse features such as "road" and "roof," evidenced by stronger activation gradients in these regions. In contrast, traditional convolution exhibits notably weaker responses to such features. This visual evidence corroborates that GCConv

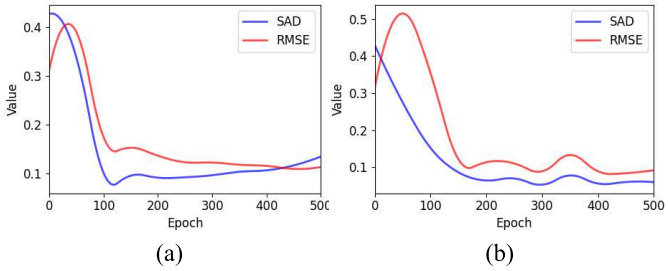


Fig. 12. Changing trends of SAD and RMSE during the training process on the Urban dataset. (a) Without recurrent learning. (b) With recurrent learning.

effectively strengthens the spatial attention module's capacity for local feature perception, particularly for fine-grained textures in HU task.

B. Integration of Recurrent Learning

Empirical analysis revealed that spectral overfitting of end-members occurs when processing HSIs with highly complex spatial features, a phenomenon particularly pronounced in Urban compared to the structurally simpler Jasper Ridge dataset. As illustrated in Fig. 12(a), initial convergence of SAD to low values failed to correspond with optimal RMSE, and further training precipitated degradation in SAD metrics—indicative of compromised generalization.

To mitigate this, we introduced recurrent spectral propagation by incorporating end-member features from the preceding iteration E_{ext}^{i-1} into the KAN learning process. This mechanism provides spectral features from the previous stage, effectively alleviating catastrophic forgetting during optimization. The results in Fig. 12(b) demonstrate substantial mitigation of overfitting, while Table VIII quantifies performance improvement on the Urban dataset.

C. KAN Substitution for MLP in Swin Transformer

As highlighted by Liu et al. [65], replacing the MLP layers in Transformers with KAN can maintain or even enhance model performance while significantly reducing parameter counts. Herein, we conduct comparative experiments to validate this hypothesis. As demonstrated in Table IX, KAN-embedded architectures achieve comparable or superior SAD and RMSE metrics compared to conventional MLP-based implementations in Swin Transformer. Notably, after replacing

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TABLE IX
COMPARISON OF KAN AND MLP IN SWIN TRANSFORMER

Dataset	Module	mSAD ±STD	mRMSE ±STD	Trainable params(M)
Jasper Ridge	MLP	0.076 ±0.026	0.088 ±0.023	60.144
	KAN	**0.047 ±0.017**	**0.070 ±0.017**	**40.225**
Urban	MLP	0.091 ±0.034	0.094 ±0.062	93.447
	KAN	**0.052 ±0.016**	**0.084 ±0.016**	**73.775**
Synthesis	MLP	0.103 ±0.068	0.114 ±0.075	91.407
	KAN	**0.039 ±0.019**	**0.045 ±0.013**	**58.085**

MLP with KAN, both mSAD and mRMSE have shown relatively significant improvements on all the datasets.

Quantitative analysis reveals that substituting MLP in the original Swin Transformer with KAN reduces trainable parameters by more than 20% relative to the original Swin Transformer architecture. This substantial reduction establishes KAN as a more efficient and lightweight module option for hyperspectral blind unmixing tasks, offering computational advantages without compromising accuracy.

VI. CONCLUSION

This article presents a novel blind HU framework that jointly infers end-members and abundances in a fully data-driven manner with only the number of end-members as input. The framework incorporates three key components: a Recurrent-KAN module for adaptive and stable end-member reconstruction, an HU-SAM adapter for global–local spatial feature fusion, and an improved Swin Transformer for parameter-efficient abundance estimation. Extensive experiments across multiple datasets demonstrate that the proposed method outperforms existing approaches, while ablation studies further validate the effectiveness of each designed component.

Despite its advantages, the framework still faces efficiency limitations, making it less suitable for real-time applications and limiting its adaptability to more complex environmental conditions. Future work will focus on developing lightweight and robust unmixing algorithms to further improve computational efficiency while preserving accuracy.

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